Sales Forecasting s Demand Intelligence Dashboard Executive Business Report

Executive Business Report

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Internship: – **XYLOFY AI** Week 3–4 Capstone Project Submission: July 2026
Technology Stack**: Python, Pandas, NumPy, Scikit-Learn, Statsmodels (SARIMA), Prophet, XGBoost, Streamlit, Plotly, Matplotlib, Git s GitHub**

Executive Summary

This project was developed to help a retail business better understand sales trends, forecast future demand, identify unusual sales patterns, and improve inventory planning. Historical sales data from 2015–2018 was analyzed using data analytics and machine learning techniques. The final system provides sales forecasts, anomaly detection, product demand segmentation, and an interactive dashboard to support business decision-making.

Key Findings

Sales Performance

Technology generated the highest revenue among all product categories.
East and West regions showed the strongest sales performance.
Average shipping time was approximately 4 days across all regions.
Sales displayed seasonal patterns with recurring peaks during January, March, October, and November.

Forecasting Results

Three forecasting models were evaluated:

Model	MAPE
SARIMA	18.97%
Prophet	21.86%
XGBoost	13.42%

XGBoost achieved the best performance and is recommended for future sales forecasting.

Three-Month Forecast

The forecasting results suggest that sales are expected to remain stable with moderate growth over the next three months.
Expected growth areas:
Technology Category
East Region
West Region
The forecast indicates continued customer demand without major declines.

Top Anomalies Detected

Two anomaly detection methods identified unusual sales periods.

Anomaly 1

Sudden sales spike likely caused by seasonal promotions or festive demand.

Anomaly 2

Unexpected increase in demand possibly linked to bulk purchases.

Anomaly 3

Temporary sales decline may be related to inventory shortages or reduced customer activity. Both anomaly detection methods identified 11 unusual periods, increasing confidence in the results.

Product Demand Segmentation
Three product demand groups were identified.
Cluster 0 – Premium Products
Examples:
Copiers
Machines
Recommended Strategy:
Maintain sufficient inventory while monitoring demand carefully.
Cluster 1 – Stable Demand Products
Examples:
Labels
Envelopes
Fasteners
Recommended Strategy:
Maintain regular stock levels with minimal excess inventory.
Cluster 2 – High Growth Products
Examples:
Phones
Chairs
Storage
Accessories
Recommended Strategy:
Prioritize inventory allocation and replenishment to avoid stock shortages.

Business Recommendations
Recommendation 1
Increase inventory levels for Technology products because they generate the highest sales revenue.
Recommendation 2
Use XGBoost forecasting results for inventory planning because it achieved the highest forecasting accuracy.
Recommendation 3
Prepare inventory before seasonal demand peaks observed during January, March, October, and November.

Risk / Limitation
The forecasting system relies on historical sales data. Unexpected market events, economic changes, or supply chain disruptions may affect future sales patterns and reduce forecast accuracy.

Result
https://roshanmanjal-sales-forecasting-app-wkipnm.streamlit.app/

Repo link:
https://github.com/roshanmanjal/Sales_Forecasting

Conclusion
The Sales Forecasting and Demand Intelligence System successfully combines forecasting, anomaly detection, and product segmentation into a single business solution. The system can help improve inventory planning, demand forecasting, and strategic decision-making while supporting future business growth.